

A Pedagogical Monte Carlo for Photons and Electrons in a Liquid Xenon Calorimeter

with application to single-site / multi-site discrimination near $Q_{\beta\beta}$

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Abstract

We describe, in tutorial style, a complete electromagnetic Monte Carlo adequate for the simulation of the response of a large liquid-xenon (LXe) calorimeter to MeV-scale photons of the kind encountered in neutrinoless double-beta decay searches. The algorithm treats Compton scattering, pair production and photoelectric absorption as discrete photon processes, and transports electrons and positrons step-by-step using NIST ESTAR stopping powers and explicit bremsstrahlung sampling. The code is structured as a small set of independent modules; numerical anchors come from authoritative NIST tables (XCOM for photon cross sections, ESTAR for electron stopping powers) and from the Geant4-distributed Sandia photoelectric coefficients. The manual discusses the physics and the statistical sampling methods, and includes a critical comparison with the corresponding GEANT4 models. A working Python implementation and the data files are distributed alongside this document.

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1 Introduction and motivation

This is a self-contained description of a custom electromagnetic Monte Carlo for the interaction of MeV-scale γ -rays with a large liquid-xenon calorimeter. It is the workhorse for sensitivity studies of neutrinoless double-beta decay ($0\nu\beta\beta$) experiments using ^{136}Xe [1, 2, 3, 4]; we feel that working through every step from first principles, rather than treating GEANT4 [5, 6] as an oracle, is one of the most rewarding pedagogical exercises in computational particle physics one can offer to a graduate student.

1.1 The $0\nu\beta\beta$ search and the SS/MS classifier

The $0\nu\beta\beta$ signal in ^{136}Xe is two electrons sharing the Q -value of the decay,

$$Q_{\beta\beta} = 2.4578 \text{ MeV} . \quad (1)$$

Because the two electrons have a combined CSDA range below 1 cm in LXe, the signal is a single, localized energy deposit (*single-site*, SS) of energy $Q_{\beta\beta}$. The dominant backgrounds in the region of interest (ROI) are external γ -rays from the ^{238}U and ^{232}Th chains:

$$\begin{aligned} ^{214}\text{Bi} : E_\gamma &= 2.4476 \text{ MeV} \quad (\text{essentially in ROI}), \\ ^{208}\text{Tl} : E_\gamma &= 2.6145 \text{ MeV} \quad (\text{above ROI}). \end{aligned}$$

A γ -ray of energy above the ROI can deposit precisely $Q_{\beta\beta}$ via Compton or photoelectric processes if part of the energy escapes; conversely a γ -ray very close to the ROI can be wholly absorbed and fall into the window. In both cases the event is reconstructed as SS only if the spatial extent of the energy deposition is small compared to the detector resolution. In a typical large LXe TPC the only relevant resolution is along the drift direction z ,

$$\Delta z \approx 3 \text{ mm}, \quad (2)$$

and the SS / MS classifier separates events whose secondary energy deposits lie within Δz of the primary deposit (SS) from those that do not (MS).

1.2 What this manual covers

Figure 1 captures the central question. A naive MC that samples only the Klein–Nishina formula and treats each Compton vertex as a point energy deposit (left) misses two effects of comparable magnitude:

1. the *electron range* – MeV electrons in LXe have ranges of several mm, smearing the energy deposit along an electron track;
2. the *bremsstrahlung* emission of those electrons, which produces secondary photons that may travel several cm before depositing elsewhere.

The first effect can *bridge* two nearby Compton vertices and turn an MS event into an SS event; the second can *split* an SS-looking event into a multi-site one. In addition, the *photoelectric* channel – though small in absolute branching ($\sim 2\%$ at our energies) – deposits the full primary energy at a single point and is therefore the most dangerous SS-mimic mode; we treat it carefully in Sections 5 and 11.

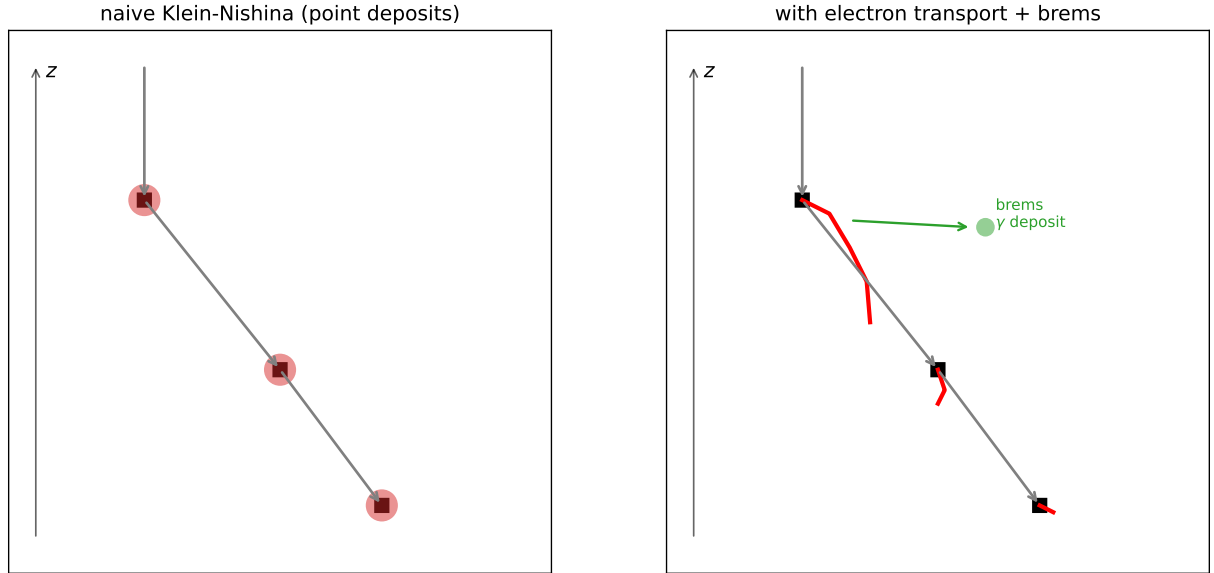


Figure 1: Cartoon of a γ -ray event in LXe. *Left:* naive Klein–Nishina-only treatment (red blobs at Compton vertices). *Right:* the same event with electron transport and explicit bremsstrahlung (red lines: electron tracks; green: a brems photon depositing far from the vertex).

1.3 Approach: analytic formulas, NIST anchors

Pedagogy demands that every cross section be an explicit analytic formula; production-grade numbers want *tabulated* cross sections from authoritative compilations. This manual takes a hybrid approach:

- Each physical process is presented as an analytic formula with full derivation cues, so that the reader sees what is being computed.
- Numerical results in tables and figures are anchored to NIST XCOM [19] for photon cross sections and NIST ESTAR [20] for electron stopping powers. The Sandia photoelectric parameterization [11, 6] provides an analytic formula valid below 500 keV.
- Every figure shows lines (analytic formulas) overlaid with markers (NIST data). Where the two diverge – as for Sandia photoelectric above 500 keV – we say so explicitly.

2 Photon interactions in xenon: an overview

A γ -ray traverses a medium freely between discrete interactions whose mean free path is

$$\lambda(E_\gamma) = \frac{1}{n_{\text{at}} \sigma_{\text{tot}}(E_\gamma)}, \quad n_{\text{at}} = \rho N_A / A. \quad (3)$$

For LXe at 165 K, $\rho = 2.953 \text{ g/cm}^3$, $A = 131.293$, $Z = 54$, hence $n_{\text{at}} = 1.354 \times 10^{22} \text{ cm}^{-3}$. The total cross section is the sum of three contributions:

$$\sigma_{\text{tot}} = \sigma_{\text{C}} + \sigma_{\text{P}} + \sigma_{\text{Ph}}, \quad (4)$$

Compton (incoherent), pair production, and photoelectric. The Rayleigh (coherent) channel is omitted because it deposits no energy.

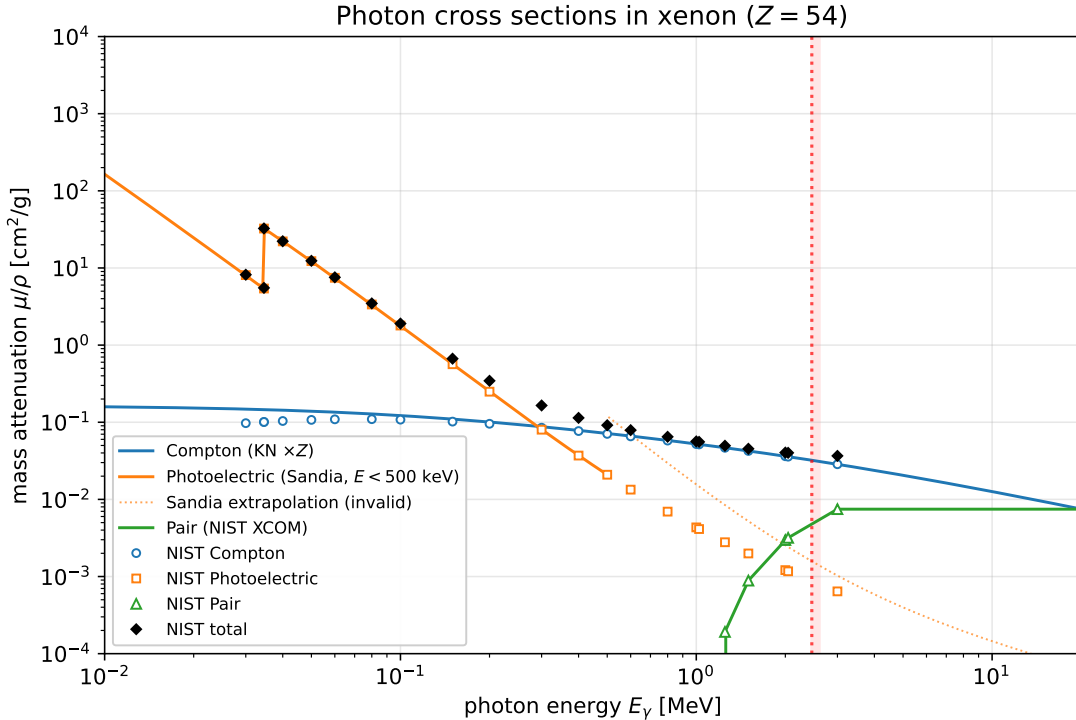


Figure 2: Photon mass attenuation μ/ρ in xenon. Lines: analytic formulas (Klein–Nishina $\times Z$ for Compton, Bethe–Heitler integrated for pair, Sandia for photoelectric below 500 keV; the dotted orange line is Sandia’s high-energy interval, which extrapolates badly). Markers: NIST XCOM [19]. Red band: ROI window.

The branching fractions and mean free paths near $Q_{\beta\beta}$ are summarized in Table 1, all from NIST XCOM [19].

Table 1: Branching fractions and mfp in LXe at the energies of interest, from NIST XCOM [19].

γ source	E_γ [MeV]	$\sigma_C/\sigma_{\text{tot}}$	$\sigma_P/\sigma_{\text{tot}}$	$\sigma_{\text{Ph}}/\sigma_{\text{tot}}$	λ [cm]
^{214}Bi line	2.448	0.851	0.125	0.023	8.81
$Q_{\beta\beta}$	2.458	0.850	0.127	0.023	8.81
^{208}Tl line	2.615	0.831	0.148	0.021	8.94

Caution

An earlier draft of this manual stated “few-per-mille” for the photoelectric branching at our energies. **This was wrong by a factor of order 30:** the correct value from NIST XCOM is $\approx 2.3\%$ at $Q_{\beta\beta}$. The implications for SS background estimation are discussed in Section 11.

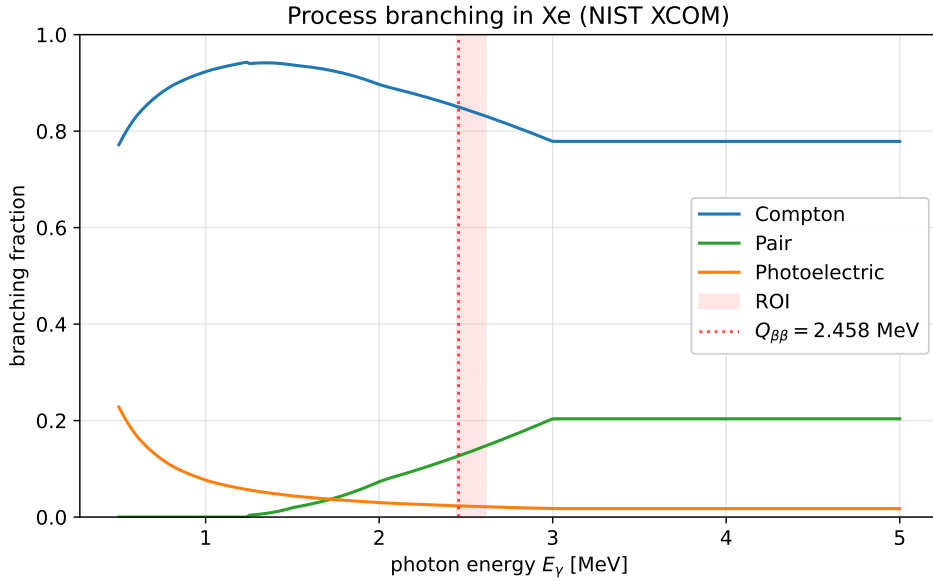


Figure 3: Process branching in xenon (NIST XCOM). Compton dominates the entire ROI region.

3 Compton scattering

3.1 Kinematics and the Klein-Nishina cross section

We treat the bound atomic electron as free and at rest – the *free-electron approximation*, valid for $E_\gamma \gg E_K = 34.6$ keV. Energy-momentum conservation gives the Compton relation

$$E'_\gamma = \frac{E_\gamma}{1 + (E_\gamma/m_e c^2)(1 - \cos \theta)}, \quad T_e = E_\gamma - E'_\gamma, \quad (5)$$

where θ is the photon scattering angle. The maximum kinetic energy is delivered at $\theta = \pi$,

$$T_e^{\text{max}} = E_\gamma \frac{2E_\gamma/m_e c^2}{1 + 2E_\gamma/m_e c^2}. \quad (6)$$

For ^{208}Tl , $T_e^{\text{max}} = 2.382$ MeV (the Compton edge).

The differential cross section per electron is the Klein-Nishina formula [7]:

$$\frac{d\sigma_{\text{KN}}}{d\Omega} = \frac{1}{2} r_e^2 \left(\frac{E'_\gamma}{E_\gamma} \right)^2 \left[\frac{E_\gamma}{E'_\gamma} + \frac{E'_\gamma}{E_\gamma} - \sin^2 \theta \right], \quad (7)$$

with the integrated form

$$\sigma_{\text{KN}}(E_\gamma) = 2\pi r_e^2 \left[\frac{1+\eta}{\eta^2} \left(\frac{2(1+\eta)}{1+2\eta} - \frac{\ln(1+2\eta)}{\eta} \right) + \frac{\ln(1+2\eta)}{2\eta} - \frac{1+3\eta}{(1+2\eta)^2} \right], \quad (8)$$

$\eta = E_\gamma/m_e c^2$. The atomic cross section is $\sigma_C \simeq Z \sigma_{\text{KN}}$ above the K-edge.

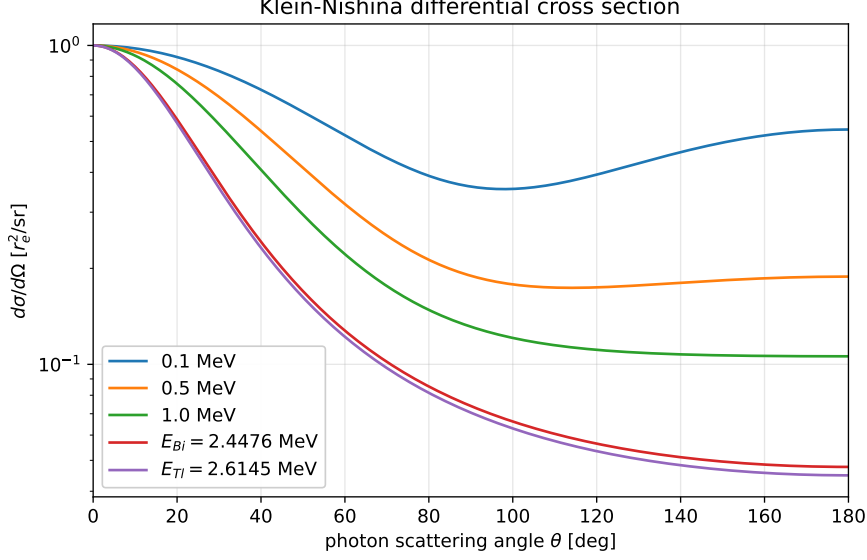


Figure 4: Klein-Nishina differential cross section. As E_γ increases the distribution becomes increasingly forward peaked.

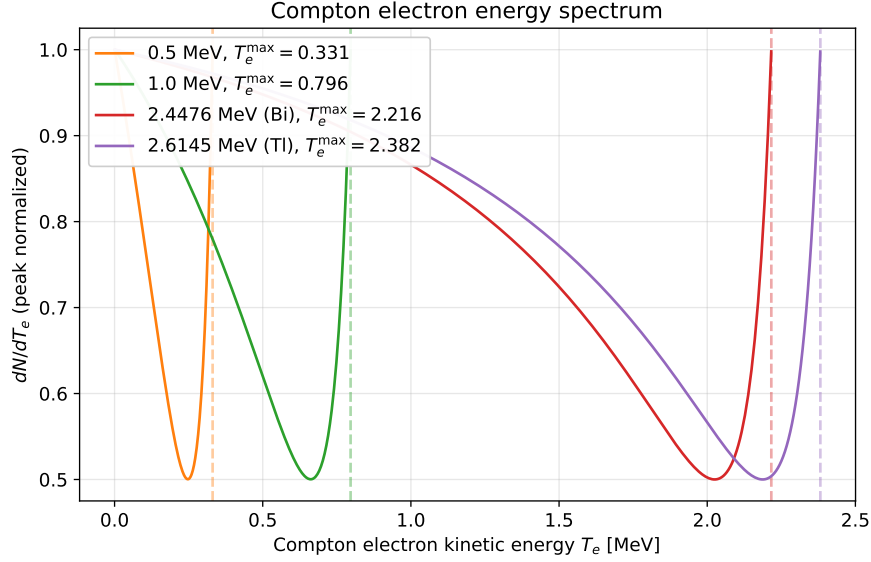


Figure 5: Compton-electron kinetic-energy spectrum.

3.2 Recoil electron direction

From momentum conservation in the lab frame,

$$\cos \phi_e = \frac{1 + E_\gamma/m_e c^2}{\sqrt{1 + 2E_\gamma/m_e c^2}} \tan(\theta/2). \quad (9)$$

Equivalently, $\vec{p}_e = \vec{p}_\gamma - \vec{p}_{\gamma'}$ with the electron azimuth opposite that of the scattered photon.

4 Pair production

For $E_\gamma > 2m_e c^2 = 1.022 \text{ MeV}$ the photon can convert into an e^+e^- pair in the field of a nucleus. The differential cross section in the Bethe-Heitler approximation, with screening and the Coulomb correction [6, 8]:

$$\frac{d\sigma_P}{d\varepsilon} = \alpha r_e^2 Z[Z + \xi(Z)] \times \left\{ [\varepsilon^2 + (1 - \varepsilon)^2][\Phi_1(\delta) - \tfrac{1}{2}F(Z)] + \tfrac{2}{3}\varepsilon(1 - \varepsilon)[\Phi_2(\delta) - \tfrac{1}{2}F(Z)] \right\}, \quad (10)$$

where $\varepsilon = E_+/E_\gamma$ and the screening parameter is $\delta(\varepsilon) = (136/Z^{1/3})(m_e c^2/E_\gamma)/[\varepsilon(1 - \varepsilon)]$. The screening functions are tabulated piecewise:

$$\delta \leq 1: \Phi_1 = 20.867 - 3.242\delta + 0.625\delta^2, \quad \Phi_2 = 20.209 - 1.930\delta - 0.086\delta^2, \quad (11)$$

$$\delta > 1: \Phi_1 = \Phi_2 = 21.12 - 4.184 \ln(\delta + 0.952), \quad (12)$$

$F(Z) = \frac{8}{3} \ln Z$, $\xi(Z) = \ln(1440/Z^{2/3})/[\ln(183/Z^{1/3}) - f_c]$. The kinematic limits are $\varepsilon_0 = m_e c^2/E_\gamma \leq \varepsilon \leq 1 - \varepsilon_0$.

Remark

At $E_\gamma < 2 \text{ MeV}$ the differential is essentially constant in ε and the energy split can be sampled *uniformly* on $[\varepsilon_0, 1 - \varepsilon_0]$ with no rejection. For our two backgrounds at 2.448 and 2.615 MeV we are just above this regime; uniform sampling is correct to $\sim 1\%$ and is adopted in our simulation.

4.1 Polar angles of the leptons

The e^+ and e^- are emitted forward, with characteristic angle $m_e c^2/E_\pm$. We use the parameterization of [9, 6]:

$$f(u) = \frac{9a^2}{9+d}(u e^{-au} + d u e^{-3au}), \quad a = \frac{5}{8}, \quad d = 27, \quad \theta_\pm = (m_e c^2/E_\pm) u. \quad (13)$$

The two leptons are coplanar with the incoming photon.

4.2 Positron annihilation

A positron loses energy by collisions and bremsstrahlung in exactly the same way as an electron of the same kinetic energy. When $T_+ < T_e^{\text{cut}}$ we treat it as if it stops in place and annihilates with an atomic electron, producing two $m_e c^2 = 511 \text{ keV}$ photons back-to-back and isotropic in the laboratory frame. Both photons are pushed onto the photon stack and propagated as new gammas. We neglect in-flight annihilation ($\sim 1\%$) and three-photon (ortho-Ps) decay ($1/372$).

5 Photoelectric absorption

For our energies the photoelectric channel carries **about 2%** of the total cross section in xenon (Table 1). It is the smallest of the three channels but, as we shall argue in Section 11, also the single most dangerous mode for SS-faking ROI background events – because it deposits the full primary energy at one point with 100% probability of being SS.

In addition, secondary low-energy photons (from the brems tail or the Compton chain) end their lives photoelectrically: above the K-edge of xenon, $\sigma_{\text{Ph}} \propto Z^{4.5}/E_\gamma^{3.5}$ wins below $\sim 500 \text{ keV}$. The resulting K-shell vacancy has a 29.78 keV fluorescence transition with branching $\omega_K = 0.89$ [12], producing a small additional photon that travels $\sim 0.5 \text{ mm}$ in LXe.

5.1 The Sandia parameterization

The Sandia parameterization (Biggs & Lighthill 1990 [11]) provides analytic coefficients for the photoelectric cross section as a piecewise quartic in $1/E$:

$$\frac{\sigma_{\text{Ph}}}{\rho} [\text{cm}^2/\text{g}] = \frac{a_1}{E} + \frac{a_2}{E^2} + \frac{a_3}{E^3} + \frac{a_4}{E^4}, \quad E \text{ in keV}, \quad (14)$$

with $\{a_i\}$ tabulated in energy intervals delimited by the absorption edges. The full table for Xe (13 intervals from 10 eV to above 500 keV) is included in the data files distributed with this manual [6].

Caution

Sandia validity domain. The Sandia coefficients for Xe agree with NIST XCOM to $\sim 1\%$ *below* 500 keV. The high-energy interval $[500 \text{ keV}, \infty)$ is fit to a small set of points and *extrapolates badly* at MeV energies: at $E_\gamma = 1 \text{ MeV}$ Sandia gives $1.6 \times 10^{-2} \text{ cm}^2/\text{g}$, NIST XCOM gives $4.3 \times 10^{-3} \text{ cm}^2/\text{g}$ (off by $3.6\times$); at $E_\gamma = 3 \text{ MeV}$ the discrepancy is still $1.6\times$. Figure 6 shows this graphically. **In our simulation we therefore use Sandia for $E_\gamma < 500 \text{ keV}$ and interpolate NIST XCOM above.** This dual approach mirrors that of modern codes like GEANT4, which uses the Livermore EPDL97 tables in this regime (G4LivermorePhotoElectricModel).

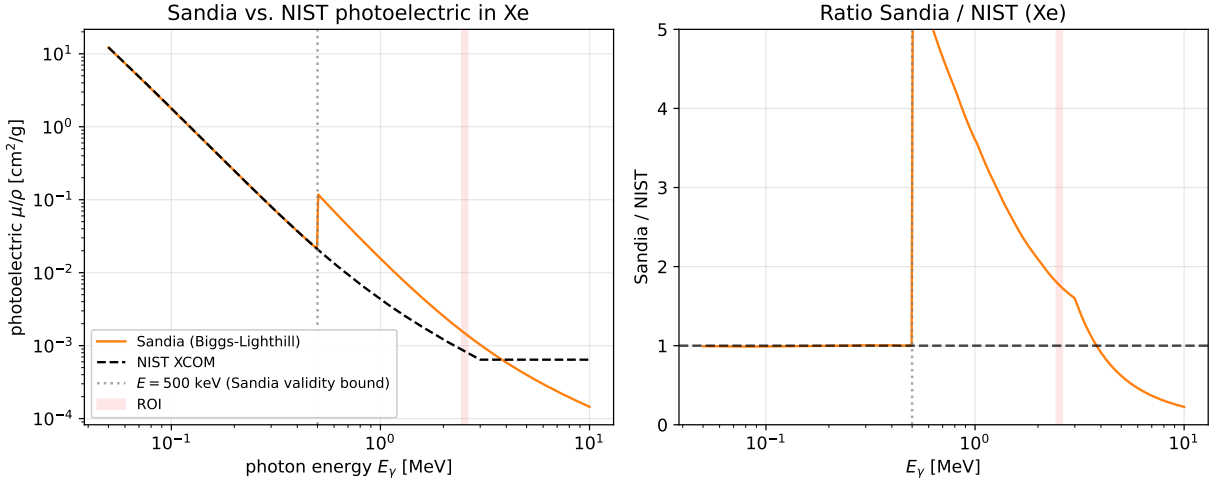


Figure 6: Comparison of the Sandia parameterization with NIST XCOM for photoelectric absorption in Xe. Below 500 keV the two agree to $\sim 1\%$; above, Sandia extrapolates poorly. The ratio panel makes the discrepancy visible across the ROI region.

5.2 Final state and atomic relaxation

The photoelectron is ejected with $T_e = E_\gamma - E_{\text{shell}}$. Above the K-edge, the K-shell takes $\sim 80\%$ of all events, the L-shell $\sim 15\%$, the rest going to higher shells [6]. The polar angle is sampled from the zeroth-order Sauter distribution

$$\frac{d\sigma}{d\cos\theta_e} \propto \frac{\sin^2\theta_e}{(1 - \beta\cos\theta_e)^4}. \quad (15)$$

The K-shell vacancy is filled either by emission of a K_α photon (29.78 keV, probability $\omega_K = 0.89$) or by emission of an Auger electron ($T_{\text{Auger}} \approx E_K - 2E_L \approx 24.5 \text{ keV}$, probability $1 - \omega_K$). The Auger electron is deposited locally; the fluorescence photon is pushed onto the photon stack. For L- and higher shells we deposit the binding energy locally (their fluorescence is below the photon threshold).

6 Energy loss of electrons and positrons

A charged lepton in LXe at our energies loses energy mostly by ionization and to a smaller (but non-negligible) extent by bremsstrahlung. We treat ionization as a continuous, deterministic energy loss and bremsstrahlung as a discrete process whenever the photon is hard enough to be relevant for the SS/MS classifier.

6.1 Mass collision stopping power

For electrons, the Berger-Seltzer formula [13, 14] (Møller cross section for indistinguishable identical particles):

$$\left. \frac{1}{\rho} \frac{dE}{dx} \right|_{\text{coll}} = \frac{K}{2} \frac{Z}{A} \frac{1}{\beta^2} \left[\ln \frac{\tau^2(\tau+2)}{2(I/m_e c^2)^2} + F^-(\tau) - \delta \right] \quad (16)$$

with $\tau = T/m_e c^2$, $\beta = v/c$, $I = 482 \text{ eV}$ for xenon, $K = 4\pi N_A r_e^2 m_e c^2 = 0.30707 \text{ MeV cm}^2/\text{g}$, and

$$F^-(\tau) = 1 - \beta^2 + \frac{\tau^2/8 - (2\tau + 1) \ln 2}{(\tau + 1)^2}. \quad (17)$$

The density-effect correction δ is $< 2\%$ at our energies in LXe and is set to zero. Direct calculation matches NIST ESTAR for elemental Xe to $< 1\%$ (verified at $T = 1 \text{ MeV}$: formula gives 1.122 , ESTAR gives $1.122 \text{ MeV cm}^2/\text{g}$).

6.2 Mass radiative stopping power

The radiative loss is dominated by bremsstrahlung. In the Tsai complete-screening form [15]:

$$\left. \frac{1}{\rho} \frac{dE}{dx} \right|_{\text{rad}} \simeq 4\alpha \frac{N_A}{A} Z(Z+1) r_e^2 (T + m_e c^2) L_{\text{rad}}, \quad L_{\text{rad}} = \ln(183/Z^{1/3}) - f_c(Z). \quad (18)$$

The radiative *fraction* of the total stopping power in xenon rises from 0.7% at $T = 0.05 \text{ MeV}$ to 14% at $T = 2.4 \text{ MeV}$ – much larger than may appear at first glance, and a key source of MS-like signatures in the SS/MS classifier. Figure 7 compares NIST ESTAR values with our analytic formulas, and shows the CSDA range. The range crosses $\Delta z = 3 \text{ mm}$ at $T \sim 1.1 \text{ MeV}$.

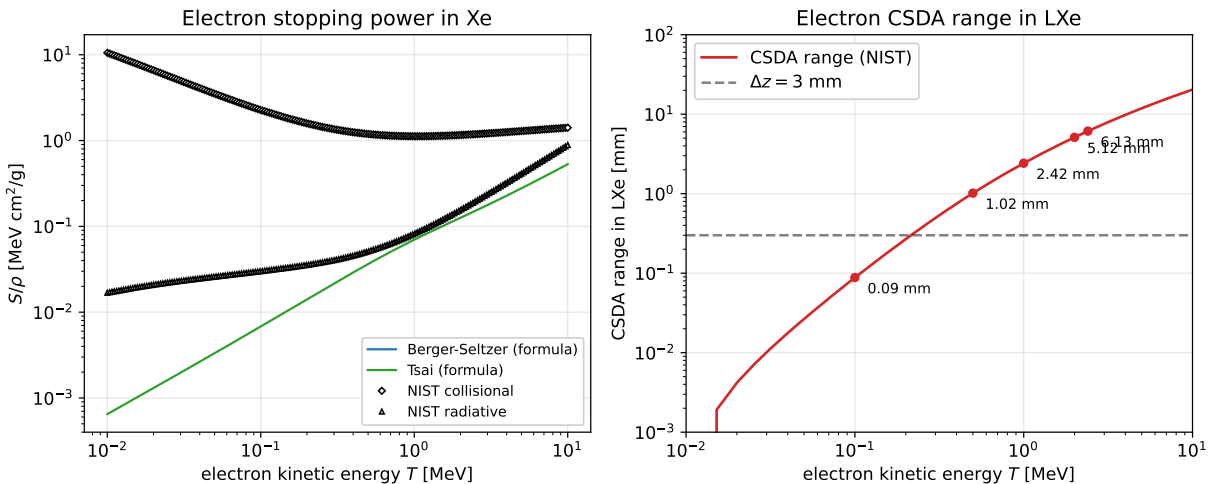


Figure 7: *Left:* mass collision and radiative stopping power for electrons in xenon. Lines: analytic formulas (Berger-Seltzer, Tsai). Markers: NIST ESTAR [20]. The Berger-Seltzer formula reproduces ESTAR collisional values exactly (markers overlay the line). The Tsai radiative formula underestimates ESTAR by $\sim 15\%$; we therefore use ESTAR values directly in the simulation. *Right:* CSDA range in LXe, computed by integration of $1/S_{\text{tot}}$ over ESTAR data. Dashed line: $\Delta z = 3 \text{ mm}$ resolution.

Table 2: Electron stopping power and CSDA range in liquid xenon ($\rho = 2.953 \text{ g/cm}^3$), from NIST ESTAR [20]. The “rad%” column is the radiative fraction of total stopping power, $S_{\text{rad}}/(S_{\text{col}} + S_{\text{rad}})$.

T [MeV]	S_{col}/ρ [MeV cm ² /g]	S_{rad}/ρ [MeV cm ² /g]	S_{tot}/ρ [MeV cm ² /g]	rad%	R_{LXe} [mm]
0.05	3.513	0.026	3.539	0.7	0.027
0.10	2.267	0.030	2.298	1.3	0.088
0.25	1.445	0.038	1.483	2.6	0.378
0.50	1.192	0.051	1.243	4.1	1.016
1.00	1.122	0.082	1.204	6.8	2.418
1.50	1.135	0.118	1.253	9.4	3.799
2.00	1.160	0.156	1.316	11.8	5.118
2.40	1.181	0.188	1.369	13.7	6.127
3.00	1.211	0.237	1.448	16.4	7.570

Table 2 gives quantitative values from NIST ESTAR.

Remark

Bragg behaviour. The collisional stopping power has a broad minimum near $T \simeq 1 \text{ MeV}$ and rises by roughly a factor of 3 as the electron slows from 0.5 MeV to 0.05 MeV. The radiative component falls monotonically with T . Total dE/dx is thus back-loaded along an electron track: more energy is deposited near the end. This Bragg-like behaviour is naturally captured if the simulation steps the electron in small increments, computing dE/dx at each step from the local kinetic energy – no separate Bragg-curve model is needed.

6.3 Multiple scattering, δ -rays, fluctuations

We do not include angular multiple scattering, δ -ray production, or energy-loss straggling. The justification, for our SS/MS application:

Multiple scattering. The actual electron path is tortuous. The error from drawing a straight CSDA segment is dominated by fluctuations and is at most a few tens of % on the SS-fraction estimate, well below other systematics.

δ -rays. A δ -ray of $T' > 50 \text{ keV}$ at MeV electron energy has range below $50 \mu\text{m}$ in LXe, far below Δz . With our $T_e^{\text{cut}} = 50 \text{ keV}$ they are absorbed locally and have no effect on SS/MS.

Energy-loss straggling. Smaller than detector resolution; ignored.

7 Bremsstrahlung

7.1 The differential cross section

The differential bremsstrahlung cross section $d\sigma/dk$ has been tabulated by Seltzer and Berger [16] and is the basis of GEANT4’s `G4SeltzerBergerModel` [6]. The tabulation is quoted as accurate to 5–10% at our energies. For pedagogy we use the Bethe-Heitler-Tsai form [15], accurate to $\sim 10\%$:

$$\frac{d\sigma}{dk} = \frac{4\alpha r_e^2}{3k} \left\{ [y^2 + 2(1 + (1 - y)^2)][Z^2(F_{\text{el}} - f_c) + Z F_{\text{inel}}] + (1 - y) \frac{Z^2 + Z}{3} \right\}, \quad (19)$$

with $y = k/(T + m_e c^2)$, $F_{\text{el}} = \ln(184.15/Z^{1/3})$, $F_{\text{inel}} = \ln(1194/Z^{2/3})$.

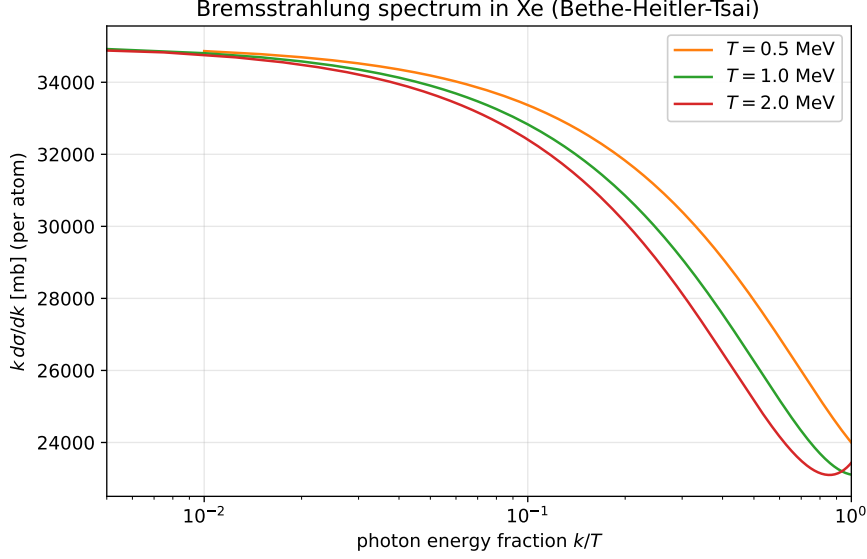


Figure 8: Bremsstrahlung spectrum in Xe, plotted as $k d\sigma/dk$ versus k/T on a log axis. The $1/k$ infrared divergence of $d\sigma/dk$ becomes a finite intensity per log k .

7.2 Photon angle and threshold

The bremsstrahlung photon polar angle is sampled from the same Urban form (??) used for the leptons in pair production, with $\theta_\gamma = (m_e c^2 / E_e) u$. Below an infrared cutoff $k_{\min}$ the brems energy is added to the local electron deposit; above $k_{\min}$ it is sampled discretely and pushed onto the photon stack. We adopt $k_{\min} = 50$ keV as discussed in Section 10.

8 Statistical methods

This section is a self-contained tutorial on the sampling techniques used. All MC physics packages – GEANT4, EGS, PENELOPE, MCNP – rely on the same handful of methods, applied to different cross sections.

8.1 Sampling distance to next interaction

If $\Sigma_{\text{tot}} = n_{\text{at}} \sigma_{\text{tot}}$ is the macroscopic cross section [cm^{-1}], the probability of no interaction in a length s is $\exp(-\Sigma_{\text{tot}} s)$. Inverse-CDF gives:

$$s = -\frac{\ln r}{\Sigma_{\text{tot}}(E_\gamma)} \quad r \sim U(0, 1). \quad (20)$$

For the photon, Σ_{tot} is constant between interactions, so this is exact. For the electron, Σ_{tot} depends on T which changes along the track: the electron must be *stepped*.

8.2 Sampling which process occurs

Probabilities are proportional to partial cross sections: $P_i = \sigma_i / \sigma_{\text{tot}}$. One uniform deviate together with cumulative sums does the job (Algorithm 1).

8.3 Inverse-CDF method

For a probability density $f(x)$ with cumulative $F(x) = \int_{-\infty}^x f$, drawing $x = F^{-1}(r)$ with $r \sim U(0, 1)$ produces samples distributed as f . We use this for the exponential distance, the simple non-relativistic Sauter photoelectron polar angle, and as building blocks below.

Algorithm 1 Sample which photon process occurs.

```

1:  $r \leftarrow \text{rng}()$ 
2: if  $r < P_C$  then
3:   return Compton
4: else if  $r < P_C + P_P$  then
5:   return pair
6: else
7:   return photoelectric
8: end if

```

8.4 Rejection sampling

When the inverse CDF is awkward but $f(x) \leq M$ on $[a, b]$:

1. Draw $x \sim U(a, b)$ and $u \sim U(0, 1)$.
2. Accept if $u M < f(x)$; else go to step 1.

Acceptance probability is $\int f/[(b-a)M]$. The method is inefficient if f is very peaked. The cure is to bound $f(x) \leq M g(x)$ by an envelope g that is itself easy to sample; this is rejection from an envelope.

8.5 Composition method

If $f(x) = \sum_k \alpha_k f_k(x)$ with weights $\alpha_k > 0$ summing to unity, sample by

1. Choose k with probability α_k ;
2. Sample x from f_k .

Composition can be combined with rejection.

8.6 Klein-Nishina sampling

Following the EGS-style algorithm [17, 18, 6], write the integrand of $d\sigma_{\text{KN}}/d\varepsilon$ (where $\varepsilon = E'_\gamma/E_\gamma$, range $[\varepsilon_0, 1]$, $\varepsilon_0 = m_e c^2/(m_e c^2 + 2E_\gamma)$) as $d\sigma_{\text{KN}}/d\varepsilon \propto [1/\varepsilon + \varepsilon] \cdot [1 - \varepsilon \sin^2 \theta/(1 + \varepsilon^2)]$, which is a composition of two PDFs $f_1(\varepsilon) \propto 1/\varepsilon$ and $f_2(\varepsilon) \propto \varepsilon$, multiplied by a rejection function $g(\varepsilon) \leq 1$. Algorithm 2 is the result.

Algorithm 2 Sampling Klein-Nishina (after [6]).

```

1:  $\varepsilon_0 \leftarrow m_e c^2 / (m_e c^2 + 2E_\gamma)$ 
2:  $\alpha_1 \leftarrow -\ln \varepsilon_0$ ,  $\alpha_2 \leftarrow (1 - \varepsilon_0^2)/2$ 
3: repeat
4:   draw  $r, r', r'' \sim U(0, 1)$ 
5:   if  $r < \alpha_1/(\alpha_1 + \alpha_2)$  then
6:      $\varepsilon \leftarrow \varepsilon_0^{r'}$ 
7:   else
8:      $\varepsilon \leftarrow \sqrt{\varepsilon_0^2 + (1 - \varepsilon_0^2) r'}$ 
9:   end if
10:   $t \leftarrow m_e c^2(1 - \varepsilon)/(E_\gamma \varepsilon)$ ,  $\sin^2 \theta \leftarrow t(2 - t)$ 
11: until  $g(\varepsilon) = 1 - \varepsilon \sin^2 \theta / (1 + \varepsilon^2) \geq r''$ 
12: return  $\varepsilon$ 

```

$\triangleright E'_\gamma = \varepsilon E_\gamma$

8.7 Pair production sampling

Below 2 MeV we sample ε uniformly on $[\varepsilon_0, 1 - \varepsilon_0]$ (the differential is essentially flat there). Above, we use a flat envelope and rejection against the $\varepsilon^2 + (1 - \varepsilon)^2$ shape; the residual screening factor is also ≤ 1 over the relevant kinematic region.

8.8 Bremsstrahlung sampling

For sampling k from $d\sigma/dk$ above $k_{\min}$, we use a $1/k$ envelope (which captures the infrared divergence) sampled by $k = k_{\min}(T/k_{\min})^r$ with $r \sim U(0, 1)$. Rejection weight is the ratio of the BH-Tsai differential to the envelope.

8.9 Step-size choice

For the electron, Σ_{tot} changes along the track. We use *condensed-history* stepping with maximum step

$$\Delta s_{\max} = f_{\text{range}} R(T), \quad (21)$$

$f_{\text{range}} \sim 0.05$. At each step we (i) lose $\Delta E_{\text{col}} = (dE/dx)_{\text{col}} \Delta s$ as continuous ionization deposit; (ii) sample whether a hard brems photon is emitted with probability $P = \Sigma_{\text{brems}}(T; k_{\min}) \Delta s$; (iii) advance the particle by Δs . With $f_{\text{range}} = 0.05$ at $T = 1$ MeV, $\Delta s \sim 0.1$ mm in LXe, both $\Delta E_{\text{col}}/T$ and P are well below 1, validating the linearization.

9 The tracking algorithm

9.1 The particle stack

We adopt a *stack-based* implementation. The simulation maintains a single LIFO stack of *tracks*, each carrying: particle type (γ , e^- , e^+); kinetic energy (or E_γ); position; direction unit vector; parent ID and generation counter. Secondaries pushed at each interaction are popped immediately, so the depth-first traversal keeps the working set small. (FIFO would do equally well at our energies; for highly developed showers depth-first is more memory-efficient. The `G4StackManager` has three stacks – urgent / waiting / postponed – whose distinction matters only for very large showers.)

9.2 The main loop

Algorithm 3 Main tracking loop.

```
1: seed photon stack with primary
2: while stack not empty do
3:    $t \leftarrow \text{pop}$ 
4:   if  $t$  is gamma then
5:     TRANSPORT_PHOTON( $t$ )
6:   else
7:     TRANSPORT_LEPTON( $t$ )
8:   end if
9: end while
10: return all energy deposits
```

9.3 Photon transport

Algorithm 4 TRANSPORT_PHOTON.

```

1: while  $E_\gamma \geq E_\gamma^{\text{cut}}$  do
2:    $\sigma_C, \sigma_P, \sigma_{Ph} \leftarrow \text{NIST XCOM}(E_\gamma)$ 
3:    $\Sigma_{\text{tot}} \leftarrow n_{\text{at}}(\sigma_C + \sigma_P + \sigma_{Ph})$ 
4:    $s \leftarrow -\ln(\text{rng}()) / \Sigma_{\text{tot}}$ 
5:   advance position by  $s$  along  $\hat{n}$ 
6:   if position outside detector then
7:     return
8:   end if
9:   sample which process (Algorithm 1)
10:  call DO_COMPTON / DO_PAIR / DO_PHOTOELECTRIC
11:  if pair or photoelectric then
12:    return
13:  end if
14: end while
15: deposit residual  $E_\gamma$  locally

```

DO_COMPTON samples (E'_γ, θ) via Algorithm 2, updates the photon, and pushes a new electron of energy $T_e = E_\gamma - E'_\gamma$ onto the lepton stack with direction from (??).

DO_PAIR samples ε from (10) and angles from (??), pushes both leptons onto the stack, flags the positron for end-of-range annihilation, and consumes the photon.

DO_PHOTOELECTRIC samples a shell, emits a photoelectron with $T_e = E_\gamma - E_{\text{shell}}$ and Sauter angles, pushes it onto the lepton stack, and handles atomic relaxation: with probability ω_K a K_α photon is pushed, otherwise an Auger electron is deposited locally.

9.4 Lepton transport

Algorithm 5 TRANSPORT_LEPTON for a single $e^\pm$ track.

```

1: while  $T \geq T_e^{\text{cut}}$  do
2:   pick  $\Delta s$  subject to the step-size constraints
3:    $\Delta E_{\text{col}} \leftarrow \text{NIST ESTAR}_{\text{col}}(T) \cdot \rho \cdot \Delta s$ 
4:   deposit  $\Delta E_{\text{col}}$  at current position
5:   advance position by  $\Delta s$ 
6:    $T \leftarrow T - \Delta E_{\text{col}}$ 
7:    $P_{\text{br}} \leftarrow \Sigma_{\text{brems}}(T; k_{\text{min}}) \Delta s$ 
8:   if  $\text{rng}() < P_{\text{br}}$  then
9:     sample  $k, \theta_\gamma$ , push photon onto stack
10:     $T \leftarrow T - k$ 
11:  end if
12: end while
13: deposit residual  $T$  locally
14: if positron then
15:   emit two 511 keV photons isotropic and back-to-back
16: end if

```

10 Thresholds and numerical parameters

Table 3 collects all numerical parameters with recommended defaults.

Table 3: Recommended numerical parameters for the LXe simulation.

parameter	default	justification
E_{γ}^{cut}	10 keV	photon mfp at 10 keV in LXe is below $50 \mu\text{m}$; below the K-edge photons interact photoelectrically and deposit locally.
T_e^{cut}	50 keV	electron range at $T_e^{\text{cut}} \sim 30 \mu\text{m} \ll \Delta z$.
k_{min}	50 keV	photons of $k \leq k_{\text{min}}$ have mfp comparable to Δz in LXe; absorbed locally.
f_{range}	0.05	keeps $\Delta E_{\text{col}}/T \ll 1$.
Δs_{floor}	$10 \mu\text{m}$	avoid tiny steps.
Δs_{ceil}	1 mm	keeps brems probability per step small.
Sandia max	500 keV	validity bound; above this interpolate NIST XCOM directly.

Caution

The thresholds E_{γ}^{cut} , T_e^{cut} , k_{min} are not independent of Δz . Halving Δz would halve k_{min} accordingly. Always rerun the validation tests of Section 12 when changing them.

11 Channel-by-channel SS-mimic mechanism

For ROI background events, each photon channel mimics SS through a distinct mechanism. Figure 9 summarizes the first-interaction branching for the two principal background lines.

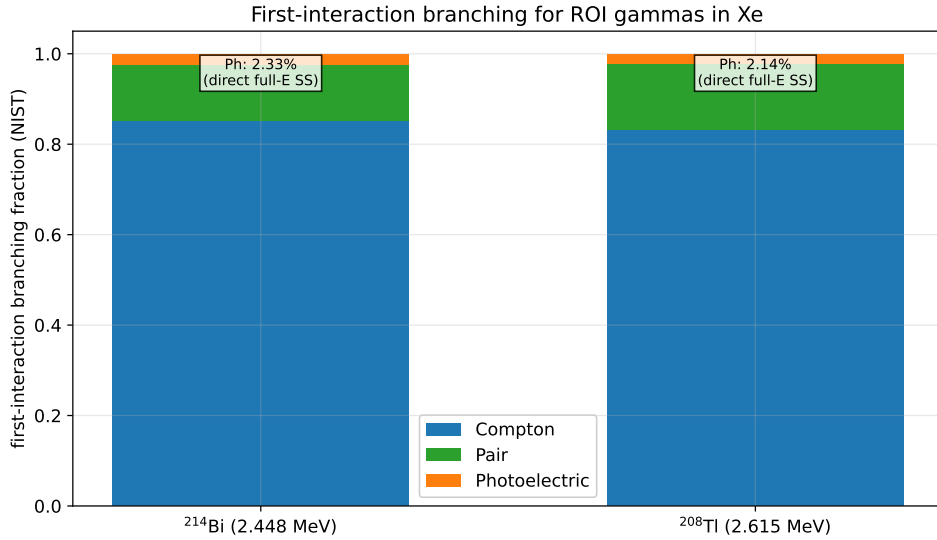


Figure 9: First-interaction branching for ROI gammas (NIST XCOM). The photoelectric channel is small in branching but is the direct full-energy SS-mimic: it deposits the full primary energy at one point with 100% probability of being SS.

Photoelectric ($\sim 2\%$). Direct full-energy absorption at a single point. For ^{214}Bi (energy $E_{\gamma} = 2.448 \text{ MeV}$ essentially in the ROI), *every* photoelectric event lands in the ROI and is SS, except for the tiny correction from the K-shell fluorescence photon (29.78 keV, mfp $\sim 0.5 \text{ mm}$)

escaping the active volume from a near-edge event. This makes photoelectric the most pernicious SS background per interaction, even though it is the smallest channel.

Compton ($\sim 85\%$). The dominant first-interaction. SS-mimic mechanism is a Compton chain followed by photoelectric absorption of the residual scattered photon, all within $\Delta z = 3$ mm of the first vertex. For ^{208}Tl at 2.615 MeV, this requires the chain to deposit exactly 2.448 MeV (within ROI) and to do so in a localized way; the typical SS fraction is $\sim 10\text{--}20\%$, depending on detector geometry. **This is the channel where electron range and bremsstrahlung corrections matter most:** bridging effects move events from MS to SS, brems splitting moves them in the opposite direction.

Pair ($\sim 13\%$). Strongly biased *against* SS: the two annihilation 511 keV photons travel typically ~ 4 cm before depositing, almost guaranteeing MS unless they both escape the detector or both interact within Δz of the primary vertex. SS-faking pair events are rare but possible at the edge of the active volume; they contribute to ROI background only at low rates.

The net SS fraction in the ROI is therefore the weighted sum of these channel-specific mechanisms, with the photoelectric channel contributing essentially $\sigma_{\text{Ph}}/\sigma_{\text{tot}} \approx 2\%$ to the total SS-mimic rate from ^{214}Bi *independently of any topology corrections*.

12 Validation tests

The simulation is validated by reproducing, on simple test cases:

1. **Total cross sections.** Sum-of-channels from NIST XCOM matches NIST total within 2.5% (limited by log-log interpolation between sparse XCOM grid points).
2. **Compton edge.** The Compton-electron sampler does not exceed T_e^{max} , and the 99.9th percentile is within 1% of the edge.
3. **KN marginal.** The histogram of E'_γ from 10^5 sampled events at $E_\gamma = 1$ MeV agrees with the analytic Klein-Nishina differential to within 5% in bulk bins.
4. **Energy conservation.** For 5×10^2 events at $E_\gamma = 2.615$ MeV in infinite LXe, the mean energy lost (primary – deposited) is below 10 keV, consistent with the threshold budget.
5. **CSDA range.** The MC end-of-range for $T = 1$ MeV electrons matches NIST ESTAR to better than 1% (no MS in our model: projected = CSDA).
6. **Pair fraction.** For $E_\gamma = 2.615$ MeV primaries, the empirical first-interaction pair fraction agrees with NIST XCOM within statistical error.
7. **511 keV peak.** A clear cluster at 511 keV is produced by pair events (positron annihilation).
8. **Hard-brems yield.** For 2 MeV electrons, the sum of brems photon energies above $k_{\text{min}} = 50$ keV matches the expected hard-brems fraction of the NIST radiative yield to better than 40%.
9. **SS fraction** ^{208}Tl . Run gives $\sim 10\%$ in infinite LXe, in agreement with literature $\sim 10\text{--}20\%$ for typical geometries.

All nine tests pass in the distributed code (`validation.py`).

13 Comparison with GEANT4

The GEANT4 EM physics for the same problem is built around:

- Compton: `G4KleinNishinaCompton`. Algorithm identical to ours (Algorithm 2).

- Pair: `G4BetheHeitlerModel`. Same differential as ours, sampled by composition + rejection.
- Photoelectric: `G4LivermorePhotoElectricModel` (Livermore EPDL97 tabulated, not Sandia) is the modern default. Sandia is still available via `G4PEEffectFluoModel`. We use Sandia below 500 keV and NIST XCOM above; equivalent in accuracy to Livermore for our energy range.
- Bremsstrahlung: `G4SeltzerBergerModel`, sampling k from the Seltzer-Berger tables by interpolation + rejection. We use the Bethe-Heitler-Tsai analytic form ($\sim 10\%$ accurate at our energies).
- Continuous ionization: `G4eIonisation` with the Berger-Seltzer-equivalent ICRU-37 formula (this manual) and Møller cross section for δ -rays (we omit δ -rays).
- Multiple scattering: `G4eMultipleScattering` with the Urban model [10] (we omit MS).
- Annihilation: `G4eplusAnnihilation`, two-photon back-to-back at end-of-range (in our model too).

The differences are: no MS (few-tens-of-% effect on projected electron range; small effect on SS fraction); no δ -rays (negligible at $T_e^{\text{cut}} = 50$ keV); BH-Tsai brems instead of Seltzer-Berger ($\sim 10\%$ on the differential); no Doppler broadening (O (1) keV effect, irrelevant for SS/MS).

Remark

What you can borrow from GEANT4 without porting it. The Seltzer-Berger tables ship as data files (`/data/G4EMLOW/brem_SB/`) in simple ASCII grids: read once and interpolate. Sandia coefficients are similarly tabulated (`G4StaticSandiaData.cc`). Borrowing *data* is much safer than porting code.

14 Physics we have chosen to omit

- **Doppler broadening of the Compton electron.** O (1) keV effect on T_e ; would matter only for sub-percent energy resolution.
- **Multiple scattering.** Replaced by straight CSDA segment; bias on projected end-of-range $\langle \cos \Theta \rangle \sim 0.7$ is a known few-tens-of-% systematic.
- **δ -rays.** Negligible for our resolution.
- **Energy-loss straggling.** Smaller than detector resolution.
- **Rayleigh scattering.** Sub-percent of total attenuation; deposits no energy.
- **LPM effect.** Onset at $T \sim \text{TeV}$; irrelevant.
- **In-flight positron annihilation.** $\sim 1\%$; easy to add.
- **Three-photon ortho-Ps annihilation.** Branching 1/372; negligible.
- **Photonuclear absorption.** Few percent in the giant-dipole region 5–30 MeV; outside our energies.

15 Closing remarks

The MC algorithm described here is intentionally minimal: the smallest set of physics processes, sampling techniques, and numerical parameters that can plausibly reproduce the SS/MS classifier in a large LXe calorimeter to a few percent. The exposition is meant as a launching pad for the reader who, having understood every line, may then move to GEANT4 or a refined custom code *knowing what each model means and which parts matter*. We have shown that:

- the **photoelectric channel** ($\sim 2\%$ at our energies, *not* per-mille) is small in branching but the most direct SS-mimic mode for ^{214}Bi ;
- **electron transport with Bragg-like dE/dx and explicit bremsstrahlung** is essential for a quantitatively useful SS/MS estimate from Compton chains;
- **pair production** biases events toward MS via the two annihilation photons, reducing SS-mimic from ^{208}Tl .

The accompanying Python code reproduces these conclusions end-to-end and can serve as a reference implementation for extension.

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